

ASC analyzer: A Python package for measuring argument structure construction usage in English texts

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Background

- In second language (L2) production research, linguistic complexity has been approached from two perspectives: (i) absolute complexity, which concerns the structural properties and interrelations of linguistic units, and (ii) relative complexity, which refers to cognitive effort, frequency of use, and statistical contingencies of those units.
- To operationalize these dimensions, recent studies have focused on argument structure constructions (ASCs), which represent how verbs combine with their arguments.
- Prior research has investigated on two major developmental trajectories:
 - L2 users advance from simpler to more complex ASCs (e.g., simple transitives → resultatives).
 - L2 users extend usage from prototypical verb–construction pairings (e.g., “give” in ditransitives) to less prototypical and more abstract combinations.
- However, a scalable and systematic tool for automatically measuring ASC usage along these trajectories remain underdeveloped.

Summary of contribution

- We present ASC Analyzer, a Python package with a RoBERTa-based tagger that automatically extracts ASCs and computes 50 indices of ASC–verb usage, including diversity, proportion, frequency, and verb–ASC associations.
- We demonstrate the tool on 6,482 L2 essays, illustrating its potential to assess L2 writing proficiency using ASC-based indices.

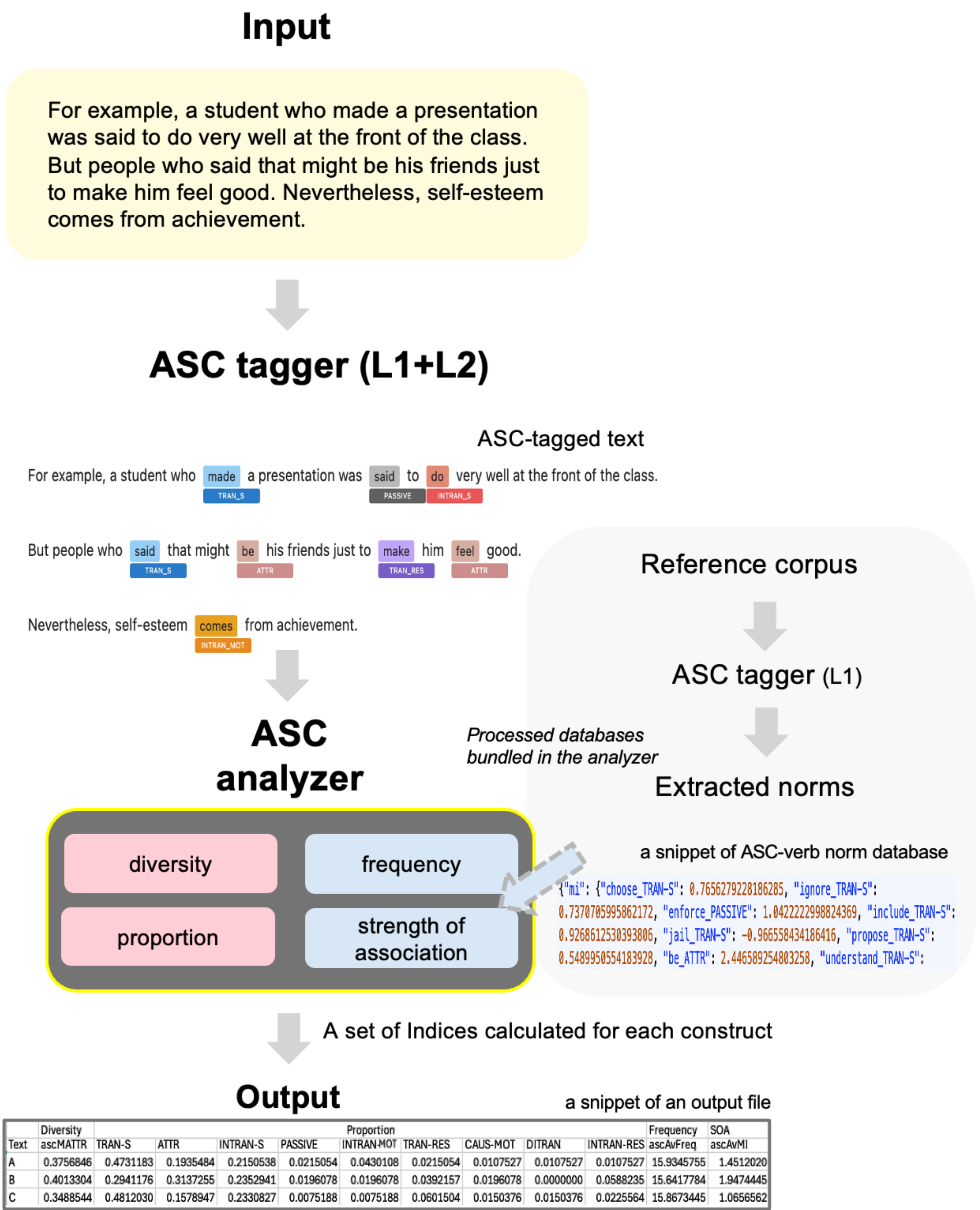
ASC-based Indices

- Diversity** Moving-average type–token ratio with sliding window w . Variants:
 - ascMATTR – ASC tokens
 - ascLemmaMATTR – ASC–verb lemma pairs
 - ascLemmaMATTRNoBe – ASC–verb lemma pairs excluding *be*
- Proportion** Share of each ASC type in a text. Variants:
 - ATTR_Prop, CAUS_MOT_Prop, DITRAN_Prop, INTRAN_S_Prop, INTRAN_MOT_Prop, INTRAN_RES_Prop, TRAN_S_Prop, TRAN_RES_Prop, PASSIVE_Prop
- Frequency** Average reference frequency of tokens from external corpora. Variants:
 - ascAvFreq – frequency of ASC tokens
 - ascLemmaAvFreq – frequency of ASC–verb pairs
- Strength of Association (SOA)** Average reference association strength between ASCs and verbs from external corpora. Metrics/Variants:
 - MI – ascAvMI, type-specific (e.g., ATTR_AvMI)
 - t -score – ascAvT, type-specific (e.g., ATTR_AvMI)
 - ΔP_{Lemma} – ascAvDeltaPLemma, type-specific (e.g., DITRAN_DeltaPLemma)
 - $\Delta P_{Structure}$ – ascAvDeltaPStructure, type-specific (e.g., CAUS_MOT_DeltaPLemma)

Reference Corpora for Norm Extraction (Frequency, SOA)

Corpus	Tokens	Sentences	ASCs Tagged	Notes
EnCOW	360,783,433	15,439,673	39,838,785	English web text (Schäfer, 2015)
SUBTLEX-US	76,965,430	5,128,462	5,665,251	8,388 subtitle files (Brysbaert et al., 2012)

ASC analyzer architecture



Application: Measuring L2 writing proficiency

- Dataset:** 6,482 L2 essays from the ELLIPSE corpus (Crossley et al., 2023)
- Composite proficiency score: average of syntax, vocabulary, phraseology, grammar sub-scores.
- Frequency & SOA norms derived from COW.

Analysis 1. Bivariate correlations

- ascMATTR showed strongest positive correlation ($r = .26$).
- ascAvFreq showed strongest negative correlation ($r = -.22$).
- Findings:** (1) Overall, more proficient writers used a wider variety of ASC types and relied less on frequent, entrenched ASC–verb pairings (except for simple transitives, using less strongly associated verbs predicted higher scores.); (2) Less frequent ASCs (e.g., intransitive resultatives, passives) were linked to higher proficiency, while simpler ASCs (e.g., attributives, simple transitives) predicted lower scores.

Analysis 2. Multivariate model

- AIC-based selection retained 12 ASC predictors.
- Final model: $R^2_{adj} = .143$, overall $r \approx .38$.
- Key positive predictors: ascMATTR, Passive (Prop & AvMI), INTRAN_RES_AvMI.
- Key negative predictors: ATTR_Prop, TRAN_S_Prop, TRAN_S_ΔP.
- Finding:** The model showed a moderate correlation with writing proficiency scores.

Analysis 3. Comparative evaluation

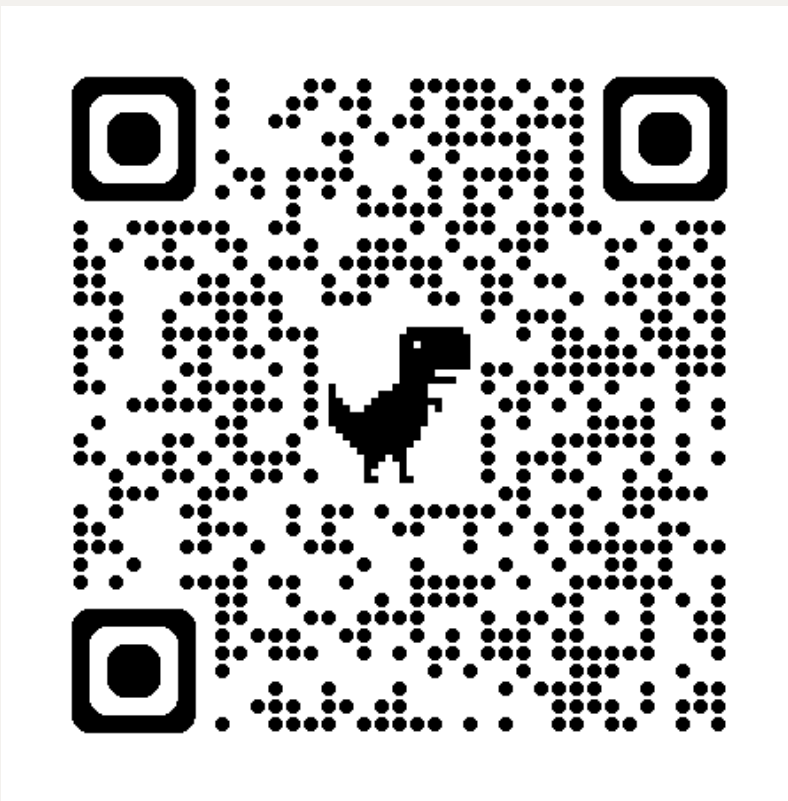
To benchmark ASC-based indices, we compared them with established complexity models in L2 research, focusing on syntactic and lexicogrammatical measures. The goal was to test whether ASCs provide greater or additional explanatory power beyond conventional indices.

- Syntactic complexity model: $R^2_{adj} = .077$.
- Lexicogrammatical model: $R^2_{adj} = .363$.
- Combined (Lexico + ASC): $R^2_{adj} = .390$ ($\Delta R^2_{adj} = .027$).
- Finding:** ASC-based indices explained more variance than syntactic measures alone and contributed additional predictive value beyond word- and phrase-level lexicogrammatical features.

Related studies

- Sung, H., & Kyle, K. (2025). Usage-based analysis of [L2 oral proficiency](#): Characteristics of argument structure construction use. *Studies in Second Language Acquisition*. [Link](#)
- Sung, H., & Kyle, K. (2025). Leveraging [pre-trained language models for linguistic analysis](#): A case of argument structure constructions. *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. [Link](#)
- Sung, H., & Kyle, K. (2024). [Annotation scheme](#) for English argument structure constructions treebank. *Proceedings of the 18th Linguistic Annotation Workshop (LAW-XVIII)*. [Link](#)
- Kyle, K., & Sung, H. (2023). An argument structure construction [treebank](#). *Proceedings of the Workshop on Construction Grammars and NLP*. [Link](#)

GitHub Repository



ASC-analyzer